

Jurabek Murodov

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EDUCATION

Stony Brook University

Expected Graduation May 2027

Bachelor of Engineering, Computer Engineering

GPA: 3.82

Relevant Coursework: Data structures, Software Techniques, Discrete Mathematics, Digital Logic Design, Digital Design Using VHDL and PLDs, Programming in C, Embedded Microcontroller Systems Design I, Electrical Circuit Analysis

SKILLS

Programming: Java, C, C++, C#, HTML, CSS, JavaScript, Python, and Arduino

Software: Visual Studio, NetBeans, Microsoft Excel, OrCad, Fusion 360, OnShape, and LTSpice

Languages: English, Tajik, Russian

PROJECTS

Intelligent Pathfinding and Memory Simulation in C

- Developed a C program for a virtual agent navigating a maze using a stack-based memory system for movement and decision-making.
- Implemented an Abstract Data Type (ADT) for the agent's actions, enabling autonomous navigation with strategies like backtracking and jumps.

Omni-Directional Car

- Developed a 3-wheel omni-directional vehicle powered by an ESP32 microcontroller, achieving advanced maneuverability and control through Bluetooth connectivity via the Dabble app.

Arduino-based Car

- Constructed a 2-wheeled autonomous vehicle with light-detection capabilities, leveraging OnShape for structural design and 3D printing technology for chassis fabrication.
- Developed adaptive navigation algorithms for autonomous path adjustment, showcasing advanced sensor use.

Marble Sorting System

- Engineered a precision sorting device with an Arduino microcontroller, incorporating two micro servos and an infrared sensor for the segregation of colored from clear marbles.
- Implemented control logic for instant sorting, showcasing advanced programming and sensor integration.

Smart USB Charger

- Designed and implemented a smart USB charger with integrated digital clock which uses a four digit seven segment display to accurately present time.
- Soldered a user-friendly interface including six LEDs that show the seconds in binary units.

Personal Website

- Utilized HTML, CSS, and JavaScript to create a personal portfolio.

EXPERIENCE

IEEE Micromouse Team

December 2024 – Present

Embedded Systems Lead

Stony Brook, NY

- Lead the development of embedded systems for the SBU Micromouse Team, integrating sensors, motor control, and decision-making algorithms.
- Collaborate with teammates to refine hardware-software interactions, optimizing navigation and control precision.
- Implement time-of-flight sensors and encoders, utilizing the VL53L0X API for accurate distance measurement and localization, improving maze-solving performance.

Stony Brook University Campus Residencies

August 2024 – Present

Resident Assistant (RA)

Stony Brook, NY

- Oversee college residence hall floor of diverse undergraduate students ranging from international to transfer students, by developing a community and ensuring each individual is comfortable in their living environment
- Plan and implement programs for residents and maintain all bulletin boards and fliers with accurate, up to date, and useful information
- Mediate roommate and suitemate conflict with the use of problem-solving and communication skills.

FTC Robotics Team

September 2022 – June 2023

Programmer

Forest Hills, NY

- Led programming for the 2022-2023 FIRST Tech Challenge, developing teleoperated and autonomous modes in Java to enhance robot performance.
- Integrated servo motors, viper slides, DC motors, and mecanum wheels for precise movement.